

FDG PDF / Org / Typst Audit

This standalone file is a starter audit comparing three artifacts:

- Published PDF: `fdg-book/fdg_book.pdf`
- Upstream Org/source material: `fdg-book/scheme/org/*.org`
- Current Typst book output: `fdg-book.pdf`

The goal is not to claim full fidelity yet. The goal is to make every partial comparison visible, so later audit runs can turn each partial or unchecked row into an exhaustive row without losing the evidence trail.

Project decisions recorded after the MVP:

- Letter layout is intentional; page-for-page reproduction of the published PDF is not a goal.
- Errata in the current book PDF is intentional project end matter.
- `typ/main.pdf` and `fdg-book.pdf` are expected to differ only by cover/index material.
- This audit should remain a standalone Typst document.
- Reference numbers should match the published PDF where possible. Current implementation seeds citation order invisibly from the published reference list, while keeping generated bibliography output.

Audit Labels

KNOWN	Evidence is concrete enough to rely on for future work.
PARTIAL	Some representative source/PDF/Typst evidence has been checked, but not the whole region.
UNCHECKED	Inventory exists, but no meaningful comparison has been completed yet.
RISK	Likely fidelity gap or known difference that needs action.
TARGET	What a later exhaustive pass should produce.

Inputs and Build Facts

Artifact	Path	Observed metadata	Audit note
Published PDF	<code>fdg-book/fdg_book.pdf</code>	249 pages; 432 x 639.36 pt; Acrobat Distiller; modified 2017-01-16.	Canonical visual/text target for book body and published index.
Current book PDF	<code>fdg-book.pdf</code>	170 pages; US letter; Typst 0.15.0; created 2026-07-03.	Generated by <code>make book</code> ; includes intentional cover/index/errata material.
Draft PDF	<code>typ/main.pdf</code>	167 pages; US letter; Typst 0.15.0; created 2026-07-03.	Generated by <code>make draft</code> ; expected to differ from <code>fdg-book.pdf</code> only by cover/index material.
Org source	<code>fdg-book/scheme/org/*.org</code>	19 Org files at top level.	<code>fdg.org</code> currently includes only preface, prologue, and chapter 1; the converter/book assembly uses the individual content files directly.
Typst source	<code>typ/content/*.typ</code>	18 content files.	Generated from Org plus repo-specific repairs and presentation support.

Reproduction commands used for this MVP:

```
pdfinfo fdg-book/fdg_book.pdf
pdfinfo fdg-book.pdf
pdfinfo typ/main.pdf
pdftotext -f 1 -l 8 fdg-book/fdg_book.pdf -
pdftotext -f 1 -l 8 fdg-book.pdf -
pdftotext -f 230 -l 249 fdg-book/fdg_book.pdf -
```

```
pdftotext -f 160 -l 170 fdg-book.pdf -
rg -n '^(|#fdg-chapter)' typ/content/*.typ
rg -n '^\\*+ ' fdg-book/scheme/org/*.org
git log --date=short --pretty=format:'%h %ad %s' -n 20
```

Executive Diff

Area	Published PDF	Org source	Current Typst PDF	MVP finding
Physical layout	249 pages, small custom page size.	No final pagination.	170 pages, letter paper.	KNOWN Letter layout is intentional. Use semantic/content checks first; layout differences are accepted unless they obscure or drop content.
Front matter	LaTeX/MIT Press title, copyright, Einstein quote, contents.	fdg.org has title metadata and quote.	Custom Typst cover, title, copyright, quote, generated contents.	PARTIAL Main matter is aligned in order; typography and contents pagination differ.
Body chapters	Preface, prologue, chapters 1-11, appendices A-C, references, index.	Matching chapter files exist, except fdg.org is not a complete driver.	Matching Typst content files exist and are included by typ/main.typ.	PARTIAL Existing notes say every chapter/appendix has at least one published-PDF spotcheck; preface, prologue, chapters 1-2, and chapter 11 had deeper passes.
End matter	References then index.	references.org and errata.org exist.	References, generated index, then errata.	KNOWN Errata is intentionally included. The generated index is intentional, but still needs content/locator audit.
References	Published extraction begins with SICP as [1].	Bib file follows published order.	Current extraction begins with SICP as [1] after citation-order seeding.	KNOWN Numbering/order now matches sampled published references. Formatting details still need exhaustive audit.
Figures	Figures embedded in published PDF.	Figure mentions in Org.	8 extracted PDF assets used in Typst.	PARTIAL Figure inventory exists; visual placement/caption fidelity is not yet exhaustive.
Code blocks	Typeset Scheme listings and results.	Org Scheme/result blocks.	Typst generated fenced Scheme blocks and result text.	PARTIAL Known converter repairs fixed misclassified

				prose and cached result placeholders in early chapters. Need systematic code/result comparison.
Math/equations	Published equations are canonical when Org is empty or drifted.	Some Org equation environments are empty or drift from PDF.	Typst contains repaired/reconstructed equations.	PARTIAL Chapter 11 equations 11.1-11.14 and 11.20-11.21 were reconstructed from PDF according to existing notes.

Structural Coverage Ledger

Region	Org -> Typst	PDF -> Typst	Current status
Preface	PARTIAL	PARTIAL	Deep pass mentioned in PDF-FIDELITY-NOTES.md; needs line-level audit entries.
Prologue	PARTIAL	PARTIAL	Deep pass mentioned; Scheme/prose classification fixes already landed.
Chapter 1	PARTIAL	PARTIAL	Deep pass mentioned; Cartan noweb placeholder and cached-result placeholder issues fixed.
Chapter 2	PARTIAL	PARTIAL	Deep pass mentioned; coordinate-independence wording/formula drift fixed.
Chapters 3-10	PARTIAL	PARTIAL	At least one PDF page spotchecked per chapter; needs exhaustive section-by-section comparison.
Chapter 11	PARTIAL	PARTIAL	Deep pass mentioned; many equations reconstructed from PDF.
Appendices A-C	PARTIAL	PARTIAL	At least one PDF page spotchecked per appendix; Appendix C end sample shows semantic match but reference-number phrasing differs.
References	PARTIAL	PARTIAL	Published order is encoded in fdg-reference-order; sampled numbering now matches. Formatting and every entry still need exhaustive audit.
Index	PARTIAL	PARTIAL	Both PDFs contain indexes, but generated entries/page numbers differ substantially.
Errata	KNOWN	KNOWN	Intentional project end matter. Compare against Org, not against the published PDF body target.

Known Findings to Preserve

These come from PDF-FIDELITY-NOTES.md and should be retained as historical audit evidence, even if future passes re-check the same regions.

- **KNOWN** 2026-07-02 spotcheck: one published-PDF page from each generated chapter and appendix was compared against Typst.
- **KNOWN** Chapter 4 equation 4.3: missing x_k subscript was restored from the published PDF.
- **KNOWN** Chapter 7 equation 7.9: punctuation was moved out of `\mathsf{g}` so the final factor renders as g , followed by punctuation.
- **KNOWN** Preface/prologue/chapter 1/chapter 2: a deeper pass fixed misclassified wrapped parenthetical prose, expanded Cartan, removed non-published cached-result placeholders, and repaired chapter 2 wording/formula drift.
- **KNOWN** Chapter 11: a deeper pass reconstructed equations 11.1-11.14 and 11.20-11.21 from the PDF because the Org source has empty equation environments there.

- **KNOWN** Chapter 11: fixes also covered Einstein, Lorentz interval terms, aligned Lorentz displays, `bar.v.double` norm notation, `make-4-tuple`, and `R^(= 1)`.
- **KNOWN** `scripts/assert-typst-regressions.mjs` preserves automated regression coverage for generated Typst structure, selected PDF text expectations, figures, labels, and bibliography keys. As of this audit pass it reports: 15 targeted files, 1 PDF text file, 18 generated files, 74 global checks, 8 figures, 599 labels, and 21 bibliography keys.

MVP Findings

A-001: Published PDF and Typst PDF use different layout targets **KNOWN**

The published PDF is 249 pages at 432 x 639.36 pt. The current book PDF is 170 pages at US letter. This means page-number comparisons are useful only inside each artifact; they are not a direct fidelity metric between artifacts.

TARGET Later exhaustive layout work should audit for dropped, obscured, or badly placed content inside the letter layout. It should not attempt to reproduce the published page geometry.

A-002: Book assembly is not the same as `fdg.org` **KNOWN**

`fdg-book/scheme/org/fdg.org` includes only:

```
#+include: preface.org :minlevel 1
#+include: prologue.org :minlevel 1
#+include: chapter001.org :minlevel 1
```

The current Typst assembly in `typ/main.typ` includes preface, prologue, chapters 1-11, appendices A-C, references, generated index, and errata. Therefore, the meaningful Org comparison target is the set of individual content files, not `fdg.org` alone.

TARGET Future converter/audit tooling should record exactly which Org file produced each Typst file, including generated or manually repaired regions.

A-003: Current PDF intentionally includes extra project end matter **KNOWN**

The published PDF end matter is references followed by index. The current Typst PDF table of contents shows references, index, and “Errata for FDG.” This is an intentional project addition, not a published-PDF fidelity bug.

TARGET Future audit rows should compare errata against `fdg-book/scheme/org/errata.org` and `typ/content/errata.typ`, while marking it out of scope for published-PDF body matching.

A-004: Reference order is seeded to match the published PDF **KNOWN**

Sampled published PDF references begin:

```
[1] Harold Abelson and Gerald Jay Sussman with Julie Sussman, Structure and Interpretation of Computer Programs
[2] Harold Abelson and Andrea deSessa, Turtle Geometry
```

Before the fix, sampled current Typst PDF references began:

```
[1] H. Abelson and A. deSessa, Turtle Geometry
[2] S. A. Papert, Mindstorms
...
[14] H. Abelson, G. J. Sussman, and J. Sussman, Structure and Interpretation of Computer Programs
```

That was a content-level difference, not just layout, because it changed citation numbers throughout the book.

The current Typst references are still generated with:

```
#bibliography("../references.bib", title: none, full: true, style: "ieee")
```

The numbering is now forced into published order by `fdg-reference-order` and `fdg-seed-bibliography-order()` in `typ/fdg-lib/refs.typ`, called from the cover/title page so it does not add pages. Verification samples from `fdg-book.pdf` after the change:

Turtle Geometry [2]
IEEE Standard for the Scheme Programming Language [10]
Spivak, Calculus on Manifolds [17]
References: [1] Structure and Interpretation of Computer Programs

TARGET Exhaustive pass should compare every citation callout and bibliography entry against the published PDF. Remaining expected differences are bibliography formatting/style details, not numbering/order.

A-005: Index exists in both PDFs but is not equivalent **PARTIAL**

The published PDF has a long index beginning after references, including symbol entries such as composition, derivative notation, tuple notation, and Scheme syntax entries. The current Typst PDF has a generated index with different entry set, sort behavior, page numbers, and visible generated-entry wording.

TARGET A full index audit should classify entries into: present in both, published-only, Typst-only, same term/different locator, and locator unavailable because page geometry intentionally differs.

A-006: Figures are inventoried but not visually audited **PARTIAL**

Current Typst uses these extracted assets:

```
fig-2-1.pdf
fig-2-2.pdf
fig-2-3.pdf
fig-4-1.pdf
fig-4-2.pdf
fig-5-1.pdf
fig-6-1.pdf
fig-7-1.pdf
```

The Org and Typst text mention the same figure numbers in the sampled regions. No exhaustive visual check has yet verified scaling, placement, caption punctuation, or page adjacency against the published PDF.

TARGET Later runs should screenshot or rasterize each figure in both PDFs, compare caption text, and mark whether visual differences are acceptable Typst styling or content loss.

A-007: Chapter 11 depends on PDF-derived repairs **KNOWN**

Chapter 11 is a special case because the Org source contains empty equation environments for a run of published equations. The Typst source now includes reconstructed equations 11.1-11.14 and 11.20-11.21, so a naive Org-vs-Typst diff will report intentional additions.

TARGET Mark PDF-derived repairs inline in generated Typst or in a machine-readable sidecar, so future converter runs do not erase them silently.

Exhaustive Audit Plan

For each region, future runs should append entries in this form:

```
ID:
Region:
Source span:
Published PDF span:
Typst span:
Status: EXACT | ACCEPTED-STYLISTIC | SOURCE-DRIFT | PDF-DERIVED-REPAIR | BUG | UNCHECKED
Evidence:
Decision:
Follow-up:
Commit(s):
```

The exhaustive audit should proceed in bounded chunks. Each chunk should update this file with concrete entries and leave any unfinished work explicitly marked **PARTIAL** or **UNCHECKED**.

Chunk	Scope	Deliverable	Notes
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1	Front matter and global rules	Cover/title/copyright/quote, contents, page numbering, draft/book differences, errata/index policy.	Mostly accepted layout/style differences. Confirm no content is dropped or obscured.
2	References and citations	All 21 bibliography entries plus all citation callouts.	High priority because references affect the whole text. Numbering/order now matches samples; formatting still needs audit.
3	Preface, prologue, chapters 1-2	First full prose/math/code/footnote workflow.	These regions already had deeper prior passes, so they are ideal for validating the audit method.
4	Chapters 3-5	Section-by-section audit, including equations, code/results, figures through 5.1.	Covers vector fields, basis fields, and integration.
5	Chapters 6-8	Equation-number audit, code/result alignment, figures 6.1 and 7.1, known chapter 7 fix verification.	Math/code density is high; keep entries section-scoped before drilling into individual blocks.
6	Chapters 9-11	Metrics, Hodge star/electrodynamics, special relativity, chapter 11 PDF-derived repairs.	Explicitly verify equations 11.1-11.14 and 11.20-11.21 against the published PDF.
7	Appendices A-C	Code/prose fidelity, citations, Appendix C tensor equation/reference checks.	Appendices are source-heavy and benefit from a separate pass.
8	Index and errata	Generated index comparison policy; errata compared against Org/Typst source.	Index page locators cannot match exactly because page geometry intentionally differs.

Recommended workflow for each chunk:

1. Extract normalized text from the published PDF, Org source, generated Typst source, and current `fdg-book.pdf`.
2. Compare first at section level, then split mismatching sections into prose, equation, code, figure, footnote, citation, and index/reference entries.
3. Classify each mismatch as `EXACT`, `ACCEPTED-STYLISTIC`, `SOURCE-DRIFT`, `PDF-DERIVED-REPAIR`, `PROJECT-ADDED`, `BUG`, or `UNCHECKED`.
4. Fix clear bugs immediately when they are local and low risk; otherwise leave an audit entry with evidence and follow-up.
5. Rebuild `fdg-book.pdf` and `typ/audit.pdf`, then record verification evidence in this file.

Settled Decisions and Remaining Questions

- Settled: letter layout is intentional.
- Settled: errata is intentional project end matter.
- Settled: `typ/main.pdf` and `fdg-book.pdf` should differ only by cover/index material.
- Settled: this audit lives as a standalone Typst document.
- Remaining: whether current generated IEEE-style bibliography formatting is acceptable now that numbering/order matches.
- Remaining: how should generated index entries be compared when page geometry intentionally differs?

Commit Trail

Commit	Date	Message
693bae8	2026-07-03	cleanup
9512cbe	2026-07-03	draft as default + makefile
413cbff	2026-07-02	fixes

a1006ad	2026-07-02	fmt
c9e554f	2026-07-02	add license, typos
650114f	2026-07-02	header
c7b3523	2026-07-02	add cover
289a9dc	2026-07-02	updates
20c5627	2026-07-02	fmt links
e12283d	2026-07-02	fmt
4eef47d	2026-07-02	fixes
eb71a72	2026-07-02	refs
8ed9337	2026-07-02	add references
4460b9f	2026-07-02	fix
3128ce4	2026-07-02	cleanup
67eefdb	2026-07-02	init
cfaf68	2026-07-02	Merge commit e1481748be83c493edfe54271d2068577aef1e2d as fdg-book
e148174	2026-07-02	Squashed fdg-book/ content from commit 9beb699
9edbac7	2026-07-02	Initial commit